Session 5
Lab Report
Contact Manager

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Course : Android App Development with Gen AI

Objective of the Contact Manager Kotlin Program

The objective of this program is to develop a simple Contact Manager application using Kotlin that demonstrates important Kotlin programming concepts such as data classes, null safety, higher-order functions, and list operations.

The program aims to create a system where contact details can be stored, managed, filtered, and displayed efficiently while ensuring safe handling of missing data like phone numbers or email addresses.

Detailed Objectives

1. To implement a Contact Manager using Kotlin
    - Store and manage contact information such as name, phone number, email, and favorite status.
2. To understand Kotlin Data Classes
    - Use a data class to represent contact information in a structured and readable format.
3. To demonstrate Kotlin Null Safety
    - Handle nullable values for phone numbers and email addresses to prevent runtime errors.
4. To implement Higher-Order Functions
    - Use functions that accept other functions as parameters to filter contacts dynamically.
5. To practice List Operations in Kotlin
    - Add, filter, and display contacts using Kotlin collection functions.
6. To improve logical thinking and reusable function design
    - Organize the program into modular functions such as adding contacts, filtering contacts, and displaying contacts.

Kotlin Code:

```
package com.example.contactmanager

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.items
import androidx.compose.material3.Card
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Text
import androidx.compose.material3.Surface
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp

// Data class
data class Contact(
    val name: String,
    val phone: String?,
    val email: String?,
    val isFavorite: Boolean
)

// Higher-order function for filtering
fun filterContacts(
    contacts: List<Contact>,
    predicate: (Contact) -> Boolean
): List<Contact> {
    return contacts.filter(predicate)
}

class MainActivity : ComponentActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        val contactList = listOf(
            Contact("john", "9876543210", "john@email.com", true),
            Contact("Diesel", null, "Diesell@email.com", false),
            Contact("Amy", "9123456780", null, true)
        )

        setContent {
            MaterialTheme {
                Surface {
                    ContactScreen(contactList)
                }
            }
        }
    }
}

@Composable
fun ContactScreen(contacts: List<Contact>) {

    val favorites = filterContacts(contacts) { it.isFavorite }

    Column(modifier = Modifier.padding(16.dp)) {

        Text(
            text = "All Contacts",
            style = MaterialTheme.typography.titleLarge
        )

        ContactList(contacts)

        Spacer(modifier = Modifier.height(20.dp))
```

```
        Text(
            text = "Favorite Contacts",
            style = MaterialTheme.typography.titleLarge
        )

        ContactList(favorites)
    }
}

@Composable
fun ContactList(contacts: List<Contact>) {

    LazyColumn {

        items(contacts) { contact ->

            Card(
                modifier = Modifier
                    .fillMaxWidth()
                    .padding(8.dp)
            ) {

                Column(modifier = Modifier.padding(12.dp)) {

                    Text(text = "Name: ${contact.name}")
                    Text(text = "Phone: ${contact.phone ?: "Not Available"}")
                    Text(text = "Email: ${contact.email ?: "Not Available"}")
                    Text(text = "Favorite: ${if (contact.isFavorite) "Yes" else "No"}")

                }
            }
        }
    }
}

@Preview(showBackground = true)
@Composable
fun ContactPreview() {

    val sampleContacts = listOf(
        Contact("john", "9876543210", "john@email.com", true),
        Contact("Diesel", null, "Diesel@email.com", false),
        Contact("Amy", "9123456780", null, true)
    )

    MaterialTheme {
        ContactScreen(sampleContacts)
    }
}
```

Minimal Explanation:

The objective of this program is to develop a Kotlin-based Contact Manager that demonstrates the use of data classes, null safety, and higher-order functions to store, filter, and display contact information efficiently.